

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

Experiment No.	14
Experiment Name	Understand Matplotlib Through Visualization

DESCRIPTION

To perform the experiment: Understand Matplotlib Through Visualization.

CODE

```
import matplotlib.pyplot as plt

values = [12, 15, 14, 10, 18, 22, 19, 17, 14, 16]
plt.figure()
plt.plot(range(1, len(values)+1), values, marker="o")
plt.title("Line Plot"); plt.xlabel("Observation"); plt.ylabel("Value")
plt.grid(True); plt.show()

plt.figure()
plt.hist(values, bins=5, edgecolor="black")
plt.title("Histogram of Values"); plt.xlabel("Value"); plt.ylabel("Frequency")
plt.show()

plt.figure()
plt.boxplot(values)
plt.title("Box Plot"); plt.ylabel("Value"); plt.show()
```

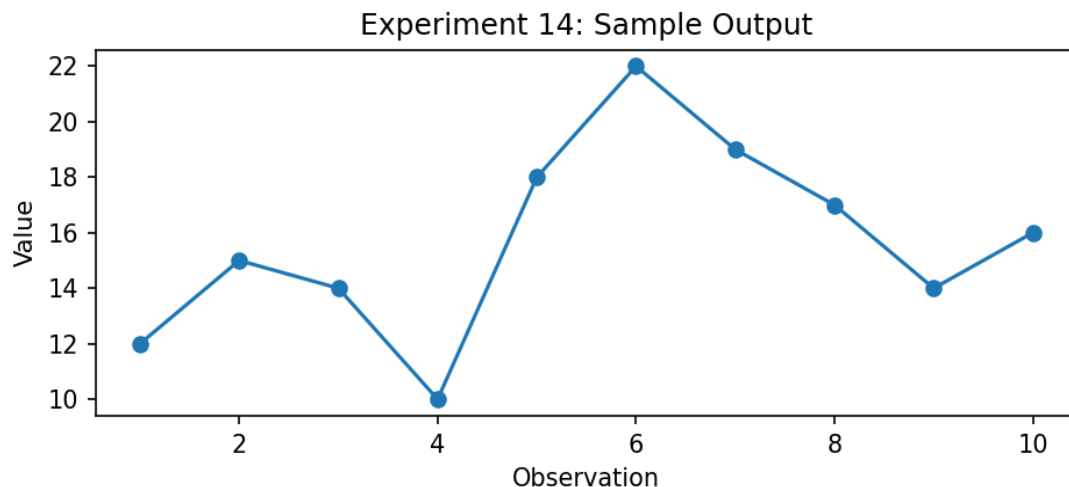
SAMPLE INPUT

Sample input: [12, 15, 14, 10, 18, 22, 19, 17, 14, 16].

OUTPUT — SAMPLE CONSOLE OUTPUT

Line plot, histogram and box plot displayed.

OUTPUT DIAGRAM / GRAPH



RESULT

The program demonstrates the stated experiment using the sample data. Verify the output by running the code in your Python environment before academic submission.

Student Signature

Faculty Signature
